

HDI Predictor Documentation

Scalability & Future Plan

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Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor - Human Development Index Classification System
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Current System Limitations

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Flask development server runs single-process by default	Limits concurrent request handling in a production setting	Medium
2	Model trained on a static, historical dataset	Predictions may drift from reality as global indicators change year over year	Medium
3	No persistent storage of past predictions	Users cannot review prediction history or track scenario comparisons over time	Low

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-process Flask dev server, suitable for demo/evaluation	Serve via Gunicorn with multiple workers behind Nginx inside the Docker image
2	Data Storage	Flat CSV + pickled model file	Move to a lightweight database (SQLite/PostgreSQL) if prediction history is added
3	Performance	Sub-second inference on a single CPU core	Add response caching for repeated identical inputs; batch-prediction endpoint for bulk country lists
4	Security	No authentication (internal evaluation tool)	Add basic auth or API keys before any public deployment

Future Roadmap

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Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Multi-country side-by-side comparison view	Q3 2025	Lets users triage several countries/regions in one screen
Phase 3	Historical trend visualization (HDI over years per country)	Q4 2025	Supports longitudinal, not just point-in-time, analysis
Phase 4	Public cloud hosting with authenticated access and prediction history	2026	Makes the tool usable by external researchers, not just local evaluators